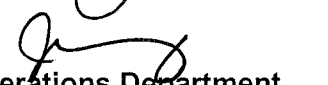


**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

STATION:	SALEM		
SYSTEM:	Reactor Coolant System		
TASK:	Measure Leakage to the Containment Sump		
TASK NUMBER:	N0020080101		
JPM NUMBER:	15-01 NRC RO A1-1		
ALTERNATE PATH:	☐	K/A NUMBER:	2.1.25
APPLICABILITY:	IMPORTANCE FACTOR:		3.9
EO ☐	RO ☒	STA ☐	SRO ☐
EVALUATION SETTING/METHOD:	Classroom		
REFERENCES:	S2.OP-AR.ZZ-0003, Overhead Annunciators Window C, Rev. 19 S2.OP-SO.RC-0004, Identifying and Measuring Leakage, Rev. 16 (Both Rev checked 8-9-16)		
TOOLS AND EQUIPMENT:	Calculator		
VALIDATED JPM COMPLETION TIME:	10 min		
TIME PERIOD IDENTIFIED FOR TIME CRITICAL STEPS:	N/A		
Developed By:	G Gauding Instructor	Date:	8-9-16
Validated By:	R Evans / N Mulford SME or Instructor	Date:	9-22-16 / 9-29-16
Approved By:	 Operations Training Manager	Date:	10/11/16
Approved By:	 Operations Department	Date:	10/11/16
ACTUAL JPM COMPLETION TIME:			
ACTUAL TIME CRITICAL COMPLETION TIME:			
PERFORMED BY:			
GRADE:	☐ SAT	☐ UNSAT	
REASON, IF UNSATISFACTORY:			
EVALUATOR'S SIGNATURE:		DATE:	

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Reactor Coolant System

TASK: Measure Leakage to the Containment Sump

TASK NUMBER: N0020080101

SIMULATOR SETUP N/A

INITIAL CONDITIONS:

Unit 2 is operating at 100% power, steady state, BOL.

At 0015, OHA C-2 CNTMT SUMP PMP START unexpectedly annunciates, along with OHA A-41. The Aux typewriter shows 21 containment sump pump has started. 2 minutes later, OHA C-2 clears, OHA A-41 annunciates, and the Aux typewriter shows 21 containment sump pump has stopped. The plant remains stable, no indications of a RCS leak are apparent, and no operations have been performed which would cause containment sump level to rise.

INITIATING CUE:

You are the RO. Calculate the leak rate into the Containment Sump by performing Section 5.3 of S2.OP-SO.RC-0004, Identifying and Measuring Leakage.

Successful Completion Criteria:

1. All critical steps completed.
2. All sequential steps completed in order.
3. All time-critical steps completed within allotted time.
4. JPM completed within validated time. Completion time may exceed the validated time if satisfactory progress is being made (and NRC concurrence is obtained).

Task Standard for Successful Completion:

1. Calculate leakage into Containment Sump of 0.85-1.0 gpm.

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____
DATE: _____

SYSTEM: Reactor Coolant System

TASK: Measure Leakage to the Containment Sump

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
		Provide blank copy of S2.OP-SO.RC-0004, and marked up copy of Attachment 1.			
*	SO.RC-4 5.3.1	<u>IF</u> either Containment Sump Pump AUTO starts, <u>THEN RECORD</u> the following on Attachment 1, Section 2.0: • Operating Containment Sump Pump Number • Time of pump start	Records <u>21 Cont Sump</u> Records time of pump start as <u>0015</u>		
*	SO.RC-4 5.3.2	CALCULATE elapsed time in minutes between start time and previous Containment Sump Pump stop time <u>AND RECORD</u> on Attachment 1, Section 2.0.	Calculates elapsed time in minutes between start time and previous Containment Sump Pump stop time and records on Attachment 1, Section 2.0 by using previous stop time of <u>1855 yesterday</u>, and with cue of <u>0015</u> today determines elapsed time is <u>320 minutes</u>.		
	SO.RC-4 5.3.3	ENSURE no draining, sampling, or liquid additions to Containment Sump have occurred during selected time frame.	Determines from initial conditions that no draining, sampling, or liquid additions to Containment Sump have occurred during selected time frame.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

NAME: _____

DATE: _____

SYSTEM: Reactor Coolant System

TASK: Measure Leakage to the Containment Sump

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
*	SO.RC-4 5.3.4	CALCULATE Containment Sump Leak Rate using Attachment 3.	Calculates Containment Sump Leak Rate using Attachment 3 (page 1 of 2), and determines the 320 minute line crosses the 0.9 gpm leak rate line. Note: Due to no increment between 300 and 350 minutes, extrapolation of data may result in leak rate slightly above or below 0.9 gpm. However, the leakrate MUST be determined to be within the bounding numbers for actions required >0.85 gpm (to initiate Section 5.6) and >1.0 gpm (to initiate S2.OP-ST.RC-0008)		
	SO.RC-4 5.3.5	RECORD Containment Sump Pump stop time and date on Attachment 1, Section 2.0.	Records Containment Sump Pump stop time as 0017 and today's date on Attachment 1, Section 2.0.		
	SO.RC-4 5.3.6	<u>IF</u> leakage to Containment Sump exceeds 1.0 gpm <u>AND</u> Unit in Modes 1-4, <u>THEN</u> : A. INITIATE S2.OP-ST.RC-0008(Q), Reactor Coolant Water Inventory Balance. B. REFER to Technical Specification 3.4.7.2.	Determines leakage to Containment Sump does not exceed 1.0 gpm		

**OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE**

NAME: _____

DATE: _____

SYSTEM: Reactor Coolant System

TASK: Measure Leakage to the Containment Sump

*	STEP NO.	STEP (*Denotes a Critical Step)	STANDARD	EVAL S/U	COMMENTS (Required for UNSAT Evaluation)
	SO.RC-4 5.3.7	IF leakage to Containment Sump exceeds 0.85 gpm, <u>THEN</u> INITIATE Section 5.6 of this procedure.	Determines leakage to Containment Sump does exceed 0.85 gpm, and initiates Section 5.6 of procedure.		
			Terminate JPM when procedure is returned to evaluator.		

OPERATOR TRAINING PROGRAM
JOB PERFORMANCE MEASURE

JOB PERFORMANCE MEASURE VALIDATION CHECKLIST

NOTE: All steps of this checklist should be performed upon initial validation. Prior to JPM usage, revalidate JPM using steps 8 and 11 below.

- W M ✓ 1. Task description and number, JPM description and number are identified.
- W M ✓ 2. Knowledge and Abilities (K/A) references are included.
- W M ✓ 3. Performance location specified. (in-plant, control room, or simulator)
- W M ✓ 4. Initial setup conditions are identified.
- W M ✓ 5. Initiating and terminating Cues are properly identified.
- W M ✓ 6. Task standards identified and verified by SME review.
- W M ✓ 7. Critical steps meet the criteria for critical steps and are identified with an asterisk (*).
- W M ✓ 8. Verify the procedure referenced by this JPM matches the most current revision of that procedure: Procedure Rev. 19 Date 9-22-16
16
- M ✓ 9. Pilot test the JPM:
a. verify Cues both verbal and visual are free of conflict, and
b. ensure performance time is accurate.
- _____ 10. If the JPM cannot be performed as written with proper responses, then revise the JPM.
- _____ 11. When JPM is revalidated, SME or Instructor sign and date JPM cover page.

SME/Instructor: [Signature]

SME/Instructor: [Signature] N. M. Ford

SME/Instructor: _____

Date: 9-22-16

Date: 9/29/16

Date: _____

INITIAL CONDITIONS:

Unit 2 is operating at 100% power, steady state, BOL.

At 0015, OHA C-2 CNTMT SUMP PMP START unexpectedly annunciates, along with OHA A-41. The Aux typewriter shows 21 containment sump pump has started. 2 minutes later, OHA C-2 clears, OHA A-41 annunciates, and the Aux typewriter shows 21 containment sump pump has stopped. The plant remains stable, no indications of a RCS leak are apparent, and no operations have been performed which would cause containment sump level to rise.

INITIATING CUE:

You are the RO. Calculate the leak rate into the Containment Sump by performing Section 5.3 of S2.OP-SO.RC-0004, Identifying and Measuring Leakage.